

FSDI: Frequency-Shaped Diffusion For Time-Series Imputation

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Abstract

In real-world web applications, especially those involving sensor networks and Internet of Things (IoT) devices, time series data are often incomplete due to network delays, device failures or logging constraints. Such missing data can severely affect downstream tasks including anomaly detection, recommendation, and A/B testing, making imputation a critical step for reliable web analytics. Diffusion models have recently achieved strong performance for time series imputation. As relevant research progresses, the spectral nature of time series has received increasing attention. However, most “frequency-aware” diffusion variants modify either the input or network architecture, but the variance schedule in the forward process remains unchanged, injecting noise with the same variance into every frequency bin. This limitation prevents diffusion from adapting to real data, where spectral energy varies irregularly across frequencies rather than following a simple high–low split. To address these issues, we propose Frequency-Shaped Diffusion (FSDI), which replaces the uniform variance schedule with a data-driven schedule in the frequency domain. Frequency bin variances are estimated from the spectral energy distribution of the data, allocated as inverses of that energy, and then Parseval-calibrated so the total noise energy exactly matches standard diffusion, preserving training stability and ensuring fair comparison. Experiments on real-world datasets demonstrate that FSDI achieves state-of-the-art performance. All code have been made publicly at <https://github.com/decisionintelligence/FSDI>.

CCS Concepts

• **Mathematics of computing** → **Time series analysis**.

Keywords

Time Series Imputation, Diffusion Model, Frequency Domain Analysis

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1 Introduction

The web has evolved into a global socio-technical infrastructure, generating continuous data streams from billions of users, devices, and services [18, 20, 26], supporting monitoring, recommendation, advertising, and online experimentation [7, 17]. However, these data streams are often incomplete due to network delays, device outages and sampling irregularities [15, 19, 52]. Such missingness propagates throughout the analytics pipeline, leading to undetected anomalies and biased models [9, 39]. These data streams are naturally analyzed as time series [21, 22, 32–34]. As a result, reliable imputation of missing time series data is essential for dependable web analytics and applications.

Research on time series imputation has evolved through several stages. Simple statistics such as mean, LOCF, or linear interpolation (lerp) provide simple baselines but ignore temporal structure. Classical statistical models including ARIMA [2], ARFIMA [13], and SARIMA [13] incorporate sequential dependence more explicitly, yet rely on strong assumptions. Machine learning methods such as MICE [42], TIDER [23], and KNNI [37] offer more flexible formulations but struggle to capture nonlinear patterns and long-range dependencies. The rise of deep learning brings models [24, 28, 35, 36] such as GRU-D [4], BRITS [3], and GAIN [48], as well as Transformer-based imputation-algorithms [8, 49], which capture nonlinear dependencies more effectively but often suffer from unstable training, or error accumulation. Most recently, diffusion models, originating with DDPM [14] and extending to time series imputation by CSDI [41], have emerged as a new generation of generative approaches [50]. Diffusion models are characterized by a forward process that progressively add noise to the original data and a learned reverse denoising process that reconstructs the original signal by approximating the inverse diffusion dynamics. This mechanism ensures stable training and strong generative capacity, making diffusion models particularly suitable for time-series imputation [12].

In addition to methodological developments, recent works have emphasized the spectral nature of time series. Frequency representations capture multi-scale components, including long-term trends,

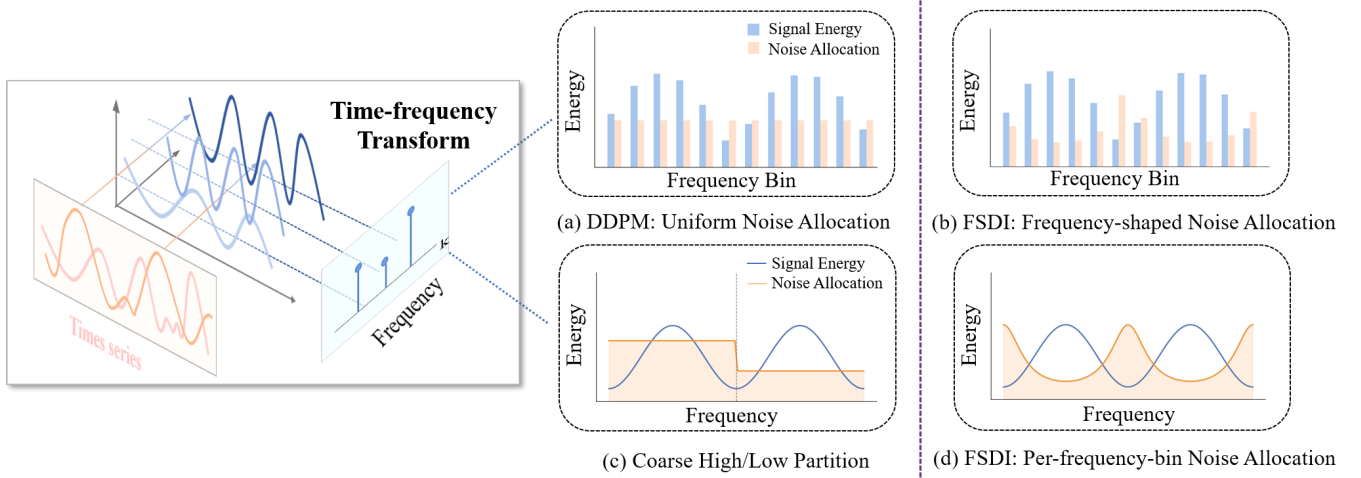


Figure 1: Challenges in frequency-shaped diffusion for imputation. (a–b) Challenge (i): Standard diffusion DDPM applies uniform noise across all frequencies, while FSDI redistributes noise according to spectral energy, preserving salient components. (c–d) Challenge (ii): Coarse spectral partitioning. Coarse high/low frequency splits fail to adapt to non-monotonic spectral distributions, whereas FSDI performs per-frequency-bin noise allocation guided by signal spectra.

seasonal peaks, and high-frequency bursts, thereby revealing where information is concentrated across the spectrum [43, 44, 51]. To exploit these properties, diffusion-based imputation algorithms have primarily incorporated frequency information on the input or network architecture [40], such as Fourier conditioning [10], high/low-frequency branches [46], or spectral decompositions prior to denoising [6]. Nevertheless, two limitations persist. (i) **Frequency-agnostic variance schedule in diffusion process.** In current designs, variance schedule remains uniform across all frequency bins, treating components of different importance identically [14, 41]. This frequency-agnostic variance schedule ignores the heterogeneous distribution of information in the frequency domain and prevents the model from explicitly prioritizing salient components, thereby limiting its ability to preserve multi-scale structure. (ii) **Coarse spectral partitioning.** Many approaches rely on high-low splits or fixed heuristics, whereas web time series exhibit non-monotonic, dataset-specific spectral coefficients in which informative bins may occur at low, mid, or high ranges [27]. Such coarse partitioning cannot adapt to these complexities and can induce systematic misallocation of noise across bins. Figure 1 provides a schematic illustration of these challenges.

To address these issues, we propose **Frequency-Shaped Diffusion for Imputation (FSDI)**, an energy-conserving reformulation of the diffusion process that makes the diffusion process spectrally aware in the frequency domain. In contrast to the uniform variance schedule that injects identical noise into all frequency bins, FSDI introduces a data-driven, per-frequency-bin variance schedule. This makes frequency awareness a property of the noising mechanism itself, rather than an add-on at the input or network architecture. **First, to adjust frequency-agnostic variance schedule in diffusion process,** we propose an energy-conserving formulation of the diffusion process that allows different frequency bins to receive

different levels of noise. By applying Parseval calibration once at initialization, the total energy remains identical to standard diffusion across all timesteps, which guarantees training stability and methodological fairness. In this way, FSDI reallocates (rather than inflates) noise across bins, ensuring that salient spectral components are not treated identically to less informative ones. **Second, to mitigate coarse spectral partitioning,** we avoid heuristic high-low splits and instead obtain data-driven characteristics directly from the data’s spectral energy distribution. Using an inverse-importance allocation, FSDI adjusts the noise variance schedule across frequency bins according to their spectral energy. Frequency bins with higher spectral energy receive less noise, while lower energy receives more. This design enables FSDI to adapt to complex, non-monotonic spectral patterns in real data without relying on coarse spectral partitioning. Taken together, these two design choices enable diffusion models to automatically perceive and exploit frequency structure while preserving the closed-form diagonal posteriors and keeping the input and network architecture unchanged, with no additional parameters introduced.

Our contributions are summarized as follows:

- We develop an energy-conserving formulation of the diffusion process that replaces the uniform variance schedule with a per-frequency-bin variance schedule. Parseval-calibrated at initialization to preserve the overall energy and closed-form posteriors.
- We design a data-driven noise allocation strategy for per-frequency-bin variance schedule, addressing frequency blindness and avoiding coarse spectral partitioning by adapting to data-driven spectra.
- We conduct experiments on multiple real-world datasets for time series imputation. FSDI achieves state-of-the-art performance across several datasets and different missing ratios, which demonstrate the effectiveness of FSDI.

2 Preliminaries

2.1 Notations and Definitions

Time series. Let $x \in \mathbb{R}^{B \times K \times L}$ denote a batch of multivariate time series, where B is the batch size, K is the number of variables, and L is the time steps. A binary mask $m \in \{0, 1\}^{B \times K \times L}$ indicates which entries are observed, with $m_{b,k,\ell} = 1$ if $x_{b,k,\ell}$ is observed and $m_{b,k,\ell} = 0$ otherwise. The binary mask $m_{\text{obs}} \in \{0, 1\}^{B \times K \times L}$ indicates which entries are available in the raw data. For conditional generation, we introduce another binary mask $m_{\text{cond}} \in \{0, 1\}^{B \times K \times L}$ that specifies which of the observed entries are revealed to the model as condition.

Transformations. We denote the unitary Discrete Fourier Transform (DFT) and its inverse (iDFT) by $\mathcal{F}_{\text{DFT}}$ and $\mathcal{F}_{\text{DFT}}^{-1}$, respectively, and write $X_{\text{DFT}} = \mathcal{F}_{\text{DFT}}(x)$. For real-valued signals we use the unitary Real Fast Fourier Transform (RFFT) and its inverse (iRFFT), denoted $\mathcal{F}_{\text{RFFT}}$ and $\mathcal{F}_{\text{RFFT}}^{-1}$. Throughout, $\mathcal{F}$ and $\mathcal{F}^{-1}$ refer to the unitary RFFT and its inverse unless stated otherwise, and we write $X = \mathcal{F}(x)$. Transforms act along the time axis. For a batch of multivariate series $x \in \mathbb{R}^{B \times K \times L}$, we obtain $X = \mathcal{F}(x) \in \mathbb{C}^{B \times K \times L_f}$, $L_f = \lfloor L/2 \rfloor + 1$. All Fourier transforms are assumed unitary. For a complex number $y = a + bi$, we use $|y| = \sqrt{a^2 + b^2}$ to denote its modulus. $\mathcal{N}_{\mathbb{C}}(0, 1)$ denotes the circularly symmetric complex Gaussian, i.e., with real and imaginary parts independently distributed as $\mathcal{N}(0, \frac{1}{2})$.

Diffusion process. We adopt the standard variance schedule $\{\beta_t\}_{t=0}^{T-1}$ with $\hat{\alpha}_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{j=0}^t \hat{\alpha}_j$, where $t \in \{0, \dots, T-1\}$ indexes diffusion steps. These parameters define the forward noising and learned reverse denoising dynamics described in later sections.

2.2 Time-frequency Transform

For real-valued time series, it is often advantageous to examine its temporal structure in the frequency domain. This is typically achieved by applying DFT and iDFT, or, for real inputs, RFFT and iRFFT.

2.2.1 DFT and iDFT. The DFT maps a batch of multivariate series $x \in \mathbb{R}^{B \times K \times L}$ into its spectral representation $X_{\text{DFT}} = \mathcal{F}_{\text{DFT}}(x) \in \mathbb{C}^{B \times K \times L}$. The unitary DFT is defined as

$$X_{\text{DFT}}[k] = \frac{1}{\sqrt{L}} \sum_{n=0}^{L-1} x[n] e^{-i2\pi kn/L}, \quad (1)$$

where each coefficient $X_{\text{DFT}}[k] \in \mathbb{C}^{B \times K \times 1}$ encodes both the amplitude and the phase of the k -th frequency bin. In particular, each coefficient can be written as $X_{\text{DFT}}[k] = |X_{\text{DFT}}[k]|e^{i\phi_k}$, where $|X_{\text{DFT}}[k]|$ represents the amplitude of the k -th frequency component and ϕ_k its phase. The amplitude indicates how much the signal oscillates at frequency bin, while the phase determines the relative time shift of that oscillation.

The iDFT reconstructs the original sequence as

$$x[n] = \frac{1}{\sqrt{L}} \sum_{k=0}^{L-1} X_{\text{DFT}}[k] e^{i2\pi kn/L}, \quad (2)$$

The unitary normalization $1/\sqrt{L}$ ensures orthonormality of the transform matrix and guarantees energy preservation (Parseval's

theorem):

$$\sum_{n=0}^{L-1} |x[n]|^2 = \sum_{k=0}^{L-1} |X_{\text{DFT}}[k]|^2. \quad (3)$$

2.2.2 From DFT to RFFT. When x is real-valued, the DFT coefficients exhibit conjugate symmetry:

$$X_{\text{DFT}}[k] = \overline{X_{\text{DFT}}[(L-k) \bmod L]}. \quad (4)$$

Hence only the first $L_f = \lfloor \frac{L}{2} \rfloor + 1$ coefficients are unique. The RFFT exploits this property: instead of producing L complex values in DFT, it maps a batch of multivariate series $x \in \mathbb{R}^{B \times K \times L}$ into a compact half-spectrum $X_{\text{RFFT}} = \mathcal{F}_{\text{RFFT}}(x) \in \mathbb{C}^{B \times K \times L_f}$.

2.2.3 Parseval's theorem in RFFT. From this point on, we adopt the convention that $X = \mathcal{F}_{\text{RFFT}}(x) \in \mathbb{C}^{B \times K \times L_f}$ denotes the unitary RFFT of a batch of multivariate series $x \in \mathbb{R}^{B \times K \times L}$. Energy preservation in RFFT can be written directly with X by introducing weights

$$w_0[f] = \begin{cases} 1, & f = 0, \\ 1, & f = L/2 \text{ if } L \text{ is even,} \\ 2, & \text{otherwise,} \end{cases}$$

so that

$$\sum_{n=0}^{L-1} |x[n]|^2 = \sum_{f=0}^{L_f-1} w_0[f] |X[f]|^2. \quad (5)$$

These weights account for the conjugate pair represented by each interior bin.

2.3 Diffusion in the Frequency Domain

In standard diffusion models, the forward process is defined in the time domain by gradually adding Gaussian noise to the data. Because the RFFT is a linear unitary transform, the same process has an equivalent formulation in the frequency domain.

2.3.1 Forward Process. Given $\hat{\alpha}_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{j=0}^t \hat{\alpha}_j$ (with $\bar{\alpha}_{-1} = 1$), the standard diffusion forward process in time domain is

$$x_t = \sqrt{\hat{\alpha}_t} x_{t-1} + \sqrt{\beta_t} \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, \mathbf{I}). \quad (6)$$

Applying the unitary RFFT to both sides of (6), we obtain $X_t = \mathcal{F}(x_t)$ and $Z_t = \mathcal{F}(\epsilon_t)$, and since $\mathcal{F}$ is unitary it preserves Gaussianity and covariance, i.e., $Z_t \sim \mathcal{N}_{\mathbb{C}}(0, \mathbf{I})$. Hence,

$$X_t = \sqrt{\hat{\alpha}_t} X_{t-1} + \sqrt{\beta_t} Z_t, \quad Z_t \sim \mathcal{N}_{\mathbb{C}}(0, \mathbf{I}). \quad (7)$$

The one-step update on frequency bin f is

$$X_t^{(f)} = \sqrt{\hat{\alpha}_t} X_{t-1}^{(f)} + \sqrt{\beta_t} Z_t^{(f)}. \quad (8)$$

Equivalently, collecting all bins yields the marginal

$$X_t \sim \mathcal{N}_{\mathbb{C}}(\sqrt{\hat{\alpha}_t} X_0, (1 - \bar{\alpha}_t) \mathbf{I}). \quad (9)$$

so iRFFT gives the standard time-domain form

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \mathcal{F}^{-1}(Z_t), \quad (10)$$

which is energy-equivalent to time-domain noising by Parseval's theorem.

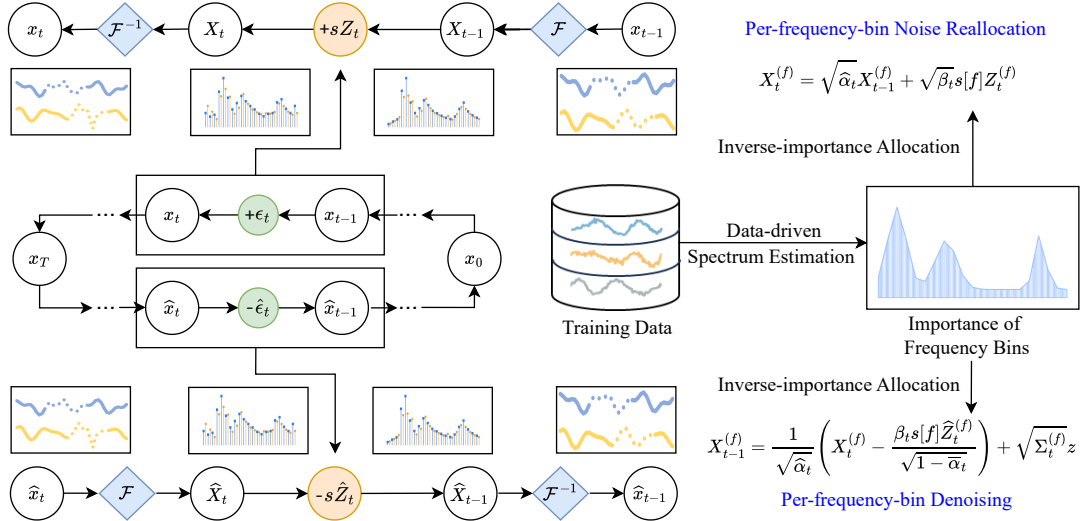


Figure 2: Frequency-shaped diffusion for time-series imputation. In the forward process, noise is reallocated per-frequency-bin using shaping coefficients derived from data-driven spectrum estimation via inverse-importance allocation (IIA). In the reverse process, per-frequency-bin denoising removes the shaped noise and maps the signal back to the time domain while preserving the overall energy.

2.3.2 Reverse Process. Under the RFFT, the forward process factorizes across frequency bins, hence the one-step posterior is diagonal in the frequency domain. For each bin f ,

$$q(X_{t-1}^{(f)} | X_t^{(f)}, X_0^{(f)}) = \mathcal{N}_C(a_t X_t^{(f)} + b_t X_0^{(f)}, \tilde{\beta}_t), \quad (11)$$

where

$$a_t = \frac{\sqrt{\tilde{\alpha}_t}(1 - \tilde{\alpha}_{t-1})}{1 - \tilde{\alpha}_t}, \quad b_t = \frac{\sqrt{\tilde{\alpha}_{t-1}}\beta_t}{1 - \tilde{\alpha}_t}, \quad \tilde{\beta}_t = \frac{1 - \tilde{\alpha}_{t-1}}{1 - \tilde{\alpha}_t} \beta_t.$$

Equivalently, in vector form over the half-spectrum,

$$q(X_{t-1} | X_t, X_0) = \mathcal{N}_C(a_t X_t + b_t X_0, \tilde{\beta}_t), \quad (12)$$

In other words, diffusion in the frequency domain is identical to the time-domain process, expressed in a different coordinate system via the unitary transform $x = \mathcal{F}^{-1}(X)$. This equivalence lets us introduce frequency-aware modeling without altering the core diffusion mechanics.

2.4 Problem Formulation

Given an incomplete time series with timestamps, we denote a sample by

$$\tilde{x} = (x, m_{\text{obs}}, T),$$

where $x \in \mathbb{R}^{B \times K \times L}$, $m_{\text{obs}} \in \{0, 1\}^{B \times K \times L}$ marks entries observed in the raw data, and $T \in \mathbb{R}^L$ is the time stamps of the time series. Probabilistic time-series imputation aims to approximate the conditional posterior $p(\tilde{x} | \tilde{x})$. Deep probabilistic methods learn an imputation function $\mathcal{M}_\theta$ that

$$\tilde{x} = \mathcal{M}_\theta(\tilde{x}),$$

where $\tilde{x} \in \mathbb{R}^{B \times K \times L}$ is the imputed time series, with $\tilde{x}_{b,i,j} = \mu_{b,i,j} \pm \sigma_{b,i,j}$ denoting the probabilistic output when $m_{b,i,j} = 1$, and $\tilde{x}_{b,i,j} = x_{b,i,j}$ otherwise.

3 Methodology

In this section, we introduce FSDI, an energy-conserving formulation of diffusion that makes the noising mechanism spectrally aware. **First**, we present a diffusion framework that supports a per-frequency-bin noise-variance schedule: each frequency bin is assigned its own variance in the forward process, and a Parseval calibration preserves the overall energy, enabling noise reallocation across frequency bins. **Second**, we introduce a data-driven schedule generation method: spectral importance is estimated directly from data, and per-frequency-bin variances are allocated inversely to that importance, thereby protecting informative components. The overall structure is shown in Figure 2.

3.1 Frequency-shaped Diffusion

In the standard diffusion process, the forward dynamics add i.i.d. Gaussian noise with the same variance across all coordinates. In the frequency domain, however, different spectral components may carry unequal semantic importance. To address this, we introduce a frequency-shaped diffusion framework that adjust the per-frequency-bin variance schedule through shaping coefficients. The specific choice of shaping coefficients will be described in Sec. 3.2.

3.1.1 Forward Process. Let $X_t = \mathcal{F}(x_t)$ denote the half-spectrum at step t , with $\hat{\alpha}_t = 1 - \beta_t$ and $\tilde{\alpha}_t = \prod_{j=0}^t \hat{\alpha}_j$ (with $\tilde{\alpha}_{-1} = 1$). An update step on frequency bin f is defined as

$$X_t^{(f)} = \sqrt{\tilde{\alpha}_t} X_{t-1}^{(f)} + \sqrt{\beta_t} s[f] Z_t^{(f)}, \quad (13)$$

where $s[f] > 0$ is a frequency-dependent shaping factor. The standard diffusion process is recovered when $s[f] = 1$. For each frequency bin, the marginal variance evolves as

$$V_t^{(f)} = \tilde{\alpha}_t V_{t-1}^{(f)} + \beta_t s^2[f], \quad V_{-1}^{(f)} = 0. \quad (14)$$

which admits the following closed-form solution:

$$V_t^{(f)} = (1 - \bar{\alpha}_t) s^2[f], \quad (15)$$

Hence the forward distribution is

$$X_t^{(f)} \sim \mathcal{N}_{\mathbb{C}}(0, V_t^{(f)}), \quad x_t = \mathcal{F}^{-1}(X_t). \quad (16)$$

As discussed earlier, our design goal is to redistribute noise across frequency bins *without changing the overall energy* compared to the time-domain diffusion. Since, under the RFFT, time-domain energy equals a *weighted* sum over the half-spectrum (Sec. 2.2), we enforce the Parseval's theorem

$$\sum_{f=0}^{L_f-1} w_0[f] V_t^{(f)} = (1 - \bar{\alpha}_t) L, \quad (17)$$

which is guaranteed for all t if the shaping coefficients are normalized once at initialization:

$$\sum_{f=0}^{L_f-1} w_0[f] s^2[f] = L.$$

In this way, frequency shaping only reallocates a fixed noise budget across bins, while the process remains energetically equivalent to the baseline, guaranteeing fair comparison and stable training.

3.1.2 Reverse Process. Given the frequency-shaped forward process, the reverse posterior remains Gaussian and factorizes across frequency bins:

$$q(X_{t-1}^{(f)} | X_t^{(f)}, X_0^{(f)}) = \mathcal{N}_{\mathbb{C}}(\mu_t^{(f)}, \Sigma_t^{(f)}), \quad (18)$$

where

$$\mu_t^{(f)} = \sqrt{\bar{\alpha}_{t-1}} \frac{\beta_t s^2[f]}{V_t^{(f)}} X_0^{(f)} + \sqrt{\hat{\alpha}_t} \frac{V_{t-1}^{(f)}}{V_t^{(f)}} X_t^{(f)}, \quad \Sigma_t^{(f)} = V_{t-1}^{(f)} \frac{\beta_t s^2[f]}{V_t^{(f)}}.$$

Thus the posterior mean is identical to the time-domain DDPM formulation, while the posterior variance is rescaled by $s^2[f]$.

Analogous to time-domain DDPM, the network predicts the *time-domain* noise $\hat{\epsilon}_t \in \mathbb{R}^{K \times L}$. Using the forward relation $X_t = \sqrt{\bar{\alpha}_t} X_0 + \sqrt{V_t} Z_t$ with $Z_t \sim \mathcal{N}_{\mathbb{C}}(0, \mathbf{I})$, we obtain the frequency-domain estimate of the noise:

$$\hat{Z}_t = \frac{\mathcal{F}(\hat{\epsilon}_t \sqrt{1 - \bar{\alpha}_t})}{\sqrt{V_t}}. \quad (19)$$

Accordingly, the practical reverse update is

$$X_{t-1}^{(f)} = \frac{1}{\sqrt{\hat{\alpha}_t}} \left(X_t^{(f)} - \frac{\beta_t s[f] \hat{Z}_t^{(f)}}{\sqrt{1 - \bar{\alpha}_t}} \right) + \sqrt{\Sigma_t^{(f)}} z, \quad z \sim \mathcal{N}_{\mathbb{C}}(0, \mathbf{I}). \quad (20)$$

3.1.3 Loss Function. Similar to CSDI, the denoising network is trained to predict the time-domain noise $\hat{\epsilon}_t$. To guide the training process, we introduce two complementary loss functions: a masked reconstruction loss in the time domain and a spectral matching loss in the frequency domain.

Algorithm 1 Training Procedure of FSDI

```

1: Input: the distribution of training data  $q(x_0)$ ; cond mask  $m_{\text{cond}}$ ,
   observed mask  $m_{\text{obs}}$ ; number of iterations  $N_{\text{iter}}$ ; variance schedule
    $\{\beta_t, \hat{\alpha}_t, \bar{\alpha}_t, V_t\}_{t=0}^{T-1}$ ; shaping coefficients  $s$  (Sec. 3.2); total de-
   noising timestamps  $T$ ; noise balance coefficient  $\lambda$ ; transforma-
   tion  $\mathcal{F}$ , frequency bin  $f = 0, \dots, L_f - 1$ .
2: Output: trained denoising function  $\epsilon_\theta$ .
3: for  $i = 1$  to  $N_{\text{iter}}$  do
4:    $t \sim \text{Uniform}\{1, \dots, T\}$ ,  $x_0 \sim q(x_0)$ 
5:    $m_{\text{tar}} = m_{\text{obs}} - m_{\text{cond}}$ ,  $x_0^{\text{co}} = x_0 \odot m_{\text{cond}}$ 
6:    $X_0 = \mathcal{F}(x_0)$ ,  $Z_t \sim \mathcal{N}_{\mathbb{C}}(0, \mathbf{I})$ 
7:   Calculate noisy targets (per frequency bin):
8:    $X_t^{(f)} = \sqrt{\bar{\alpha}_t} X_0^{(f)} + \sqrt{1 - \bar{\alpha}_t} s[f] Z_t^{(f)}$ 
9:    $x_t = \mathcal{F}^{-1}(X_t)$ 
10:  Predict time-domain noise:  $\hat{\epsilon}_t \leftarrow \epsilon_\theta(x_t, x_0^{\text{co}}, t)$ .
11:  Calculate true time-domain noise:
12:   $\epsilon_t = \mathcal{F}^{-1}(\sqrt{V_t} Z_t) / \sqrt{1 - \bar{\alpha}_t}$ 
13:  Calculate time-domain loss  $\mathcal{L}_{\text{time}} = \mathbb{E} \|(\hat{\epsilon}_t - \epsilon_t) \odot m_{\text{tar}}\|^2$ 
14:  Calculate frequency-domain loss  $\mathcal{L}_{\text{freq}} = \mathbb{E} \|\hat{Z}_t - Z_t\|^2$ 
15:  Take gradient descent step on  $\nabla_\theta [\mathcal{L}_{\text{time}} + \lambda_{\text{freq}} \mathcal{L}_{\text{freq}}]$ 
16: end for
17: return  $\epsilon_\theta$ 

```

Algorithm 2 Sampling Procedure of FSDI

```

1: Input: observed data  $x_0$ ; cond mask  $m_{\text{cond}}$ ; trained denoising
   function  $\epsilon_\theta$ ; variance schedule  $\{\beta_t, \hat{\alpha}_t, \bar{\alpha}_t, V_t\}_{t=0}^{T-1}$ ; shaping coef-
   ficients  $s$ ; transformation  $\mathcal{F}$ .
2: Output: imputed sample  $\hat{x}_0$ .
3: Initialize:  $x_T = \mathcal{F}^{-1}(s \odot Z)$  with  $Z \sim \mathcal{N}_{\mathbb{C}}(0, \mathbf{I})$ .
4:  $x_0^{\text{co}} = x_0 \odot m_{\text{cond}}$ 
5: for  $t = T$  to  $1$  do
6:    $\hat{\epsilon}_t \leftarrow \epsilon_\theta(x_t, x_0^{\text{co}}, t)$ 
7:    $\hat{Z}_t = \frac{\mathcal{F}(\hat{\epsilon}_t \sqrt{1 - \bar{\alpha}_t})}{\sqrt{V_t}}$ ,  $X_t = \mathcal{F}(x_t)$ 
8:   Sample  $\hat{X}_{t-1}$  using Equation 20
9:   Invert to time domain:  $\hat{x}_{t-1} = \mathcal{F}^{-1}(\hat{X}_{t-1})$ .
10: end for
11: return  $\hat{x}_0$ 

```

Time-domain Loss. In the forward process, each noisy data X_t is constructed from the clean sequence X_0 by injecting noise Z_t in the frequency domain. Therefore, the corresponding ground-truth noise in the time domain can be recovered from Z_t

$$\epsilon_t = \frac{\mathcal{F}^{-1}(\sqrt{V_t} Z_t)}{\sqrt{1 - \bar{\alpha}_t}}. \quad (21)$$

With target mask $m_{\text{tar}} = m_{\text{obs}} - m_{\text{cond}}$, the network is only supervised at positions that are observed but not conditioned. The masked mean-squared error is

$$\mathcal{L}_{\text{time}} = \mathbb{E} \|(\hat{\epsilon}_t - \epsilon_t^{\text{true}}) \odot m_{\text{tar}}\|^2. \quad (22)$$

Frequency-domain loss. To further regularize spectral consistency, we transform the prediction into frequency domain according to Equation 19 and compare it against the true noise Z_t used in the forward process. The discrepancy is measured by the squared error

between real and imaginary parts:

$$\mathcal{L}_{\text{freq}} = \mathbb{E}(|\widehat{Z}_t - Z_t|^2). \quad (23)$$

Final objective. The overall training objective combines the time-domain loss and the frequency-domain loss::

$$\mathcal{L} = \mathcal{L}_{\text{time}} + \lambda_{\text{freq}} \mathcal{L}_{\text{freq}}, \quad \lambda_{\text{freq}} > 0. \quad (24)$$

The $\mathcal{L}_{\text{time}}$ directly optimizes the imputation accuracy under the observed mask, while the $\mathcal{L}_{\text{freq}}$ provides an auxiliary regularization that promotes consistency with the real spectral distribution. Here, λ_{freq} controls the strength of spectral regularization: setting $\lambda_{\text{freq}}=0$ recovers the CSDI objective, while a small positive value encourages the prediction to match the true noise spectral. Based the parameter sensitivity analysis in Sec. 4.4, we finally adopt $\lambda_{\text{freq}}=0.05$ as the default. The detailed training and sampling algorithm is presented in Alg.1 and 2.

3.2 Frequency Importance Estimation

We estimate frequency importance from data and convert it into a shaping coefficients. The procedure has two stages: (i) estimating a robust power spectrum from observed sequences, and (ii) allocating the noise budget by an inverse-importance allocation (IIA).

3.2.1 Data-driven Spectrum Estimation. To evaluate which frequencies carry more information, we estimate the distribution of signal energy across frequency bands. Concretely, we apply RFFT to the multivariate time series $x \in \mathbb{R}^{B \times K \times L}$, yielding the spectral representation $X \in \mathbb{C}^{B \times K \times L_f}$. For each frequency bin f , we compute the squared magnitude of the Fourier coefficients

$$P(f) = |X^{(f)}|^2, \quad (25)$$

which corresponds to the *periodogram*, i.e., an empirical estimator of the power spectral density (PSD). Intuitively, $P(f)$ tells us how much of the signal is explained by oscillations at frequency f : large values mean the data contains strong patterns at that frequency.

Since the observed sequences may contain many missing entries, a direct computation of $P(f)$ would be biased. We therefore normalize the spectrum by the effective number of observations, average across variables and samples, and apply a mild smoothing to stabilize the estimates. Finally, we normalize the resulting spectrum to have unit mean:

$$w(f) = \frac{P(f)}{\frac{1}{L_f} \sum_{u=0}^{L_f-1} P(u)}. \quad (26)$$

The frequency-importance coefficients $w(f)$ captures the *relative importance* of each frequency: higher $w(f)$ corresponds to more dominant components in the data, which we protect by injecting less noise in the diffusion process.

3.2.2 Noise Allocation by IIA. Given the frequency-importance coefficients $w(f)$, we need to determine the per-frequency noise variance $s(f)$ used in the forward diffusion. Our goal is to inject *less noise* into more important frequencies (larger $w(f)$) while maintaining the same overall energy as the standard diffusion. To this end, we adopt an inverse-importance allocation (IIA) strategy consisting of three steps.

Inverse-importance allocation. We first assign noise variance inversely proportional to the importance:

$$\hat{s}^2(f) = \frac{w(f)^{-1}}{\sum_{u=0}^{L_f-1} w(u)^{-1}} L_f, \quad (27)$$

so that bins with larger importance $w(f)$ receive proportionally less noise.

Parseval calibration. Because the time-domain energy is a weighted sum of half-spectrum variances (Sec. 2.2), we apply a final rescaling so that the total energy budget matches the standard time-domain diffusion:

$$s^2(f) \leftarrow \hat{s}^2(f) \cdot \frac{L}{\sum_{u=0}^{L_f-1} w_0[u] s^2[u]}, \quad \sum_{f=0}^{L_f-1} w_0[f] s^2[f] = L, \quad (28)$$

where $w_0(f)$ is the Parseval weights defined in Sec. 2.2. This step guarantees that frequency shaping only redistributes noise across frequencies, while the overall time-domain energy remains identical to the standard diffusion.

4 Experiments

4.1 Experimental Details

4.1.1 Experimental Setting. We implement FSDI by modifying the official CSDI codebase [41] and conduct all experiments within this framework. The training and sampling procedures follow Algorithm 1 and Algorithm 2, respectively. We use the Adam optimizer [16] with an initial learning rate of 1×10^{-3} coupled with a multi-step decay schedule. All experiments are conducted using PyTorch [31] in Python 3.9 and executed on an NVIDIA RTX 3090 GPU. The hyperparameters in our experiments is detailed in Appendix.A.1.

4.1.2 Datasets. We conduct experiments on two commonly used multivariate time series datasets. The first is the PhysioNet Challenge 2012 dataset [38], which contains clinical records from intensive care units (ICU). It includes 4000 time series with 35 physiological variables over a 48-hour window. Following prior work [3, 41], we resample the series to an hourly grid, resulting in sequences of length 48. The processed dataset is highly sparse, with approximately 80% of missing values. As the dataset lacks ground-truth, we randomly choose 10/50/90% of observed values as ground-truth on the test dataset. The second dataset is the Air Quality [47], which consists of PM2.5 measurements collected from 36 monitoring stations in Beijing over one year. Each sequence is organized into 36 time steps, and the dataset has about 13% missing values. These two datasets represent distinct domains with different missingness levels and structures, making them suitable for evaluating the imputation algorithms.

4.1.3 Baselines. We compare FSDI with a broad set of baselines in time-series imputation, including simple statistical methods (Mean and lerp), deep learning methods (BRITS [3], SAITS [8], TimesNet [43]), deep generative models (GP-VAE [11], SSGAN [29], VRIN [30], CSDI [41], CSBI [5], SSSD [1], TS-Diff [23], D³M [45], LSCD [10]) and time-series foundation model (ModernTCN [25]). These baselines provide a comprehensive comparison against which the effectiveness of FSDI can be assessed.

Table 1: Imputation performance (MAE and RMSE) on PhysioNet and Air Quality. Best results are in bold; second-best are underlined.

Method	PhysioNet 10%		PhysioNet 50%		PhysioNet 90%		Air Quality	
	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE
Mean	0.714	1.035	0.711	1.091	0.710	1.097	50.685	66.558
Lerp	0.372	0.708	0.417	0.840	0.565	0.993	15.363	27.658
BRITS	0.278	0.693	0.385	0.833	0.560	0.975	16.519	26.775
GP-VAE	0.469	0.783	0.521	0.907	0.642	1.038	23.941	40.586
SSGAN	0.323	0.662	0.449	0.852	0.670	1.060	32.999	48.951
TimesNet	0.375	0.690	0.453	0.840	0.642	1.031	22.685	39.336
CSDI	0.219	0.545	0.307	0.672	0.481	0.834	9.670	19.093
CSBI	0.230	0.547	0.310	0.649	–	0.834	9.800	19.000
SAITS	0.232	0.583	0.315	0.735	0.565	0.971	15.424	30.558
ModernTCN	0.351	0.697	0.440	0.803	0.647	1.026	24.089	40.052
LSCD	<u>0.211</u>	<u>0.494</u>	<u>0.303</u>	0.664	<u>0.479</u>	<u>0.832</u>	9.069	17.914
FSDI (Ours)	0.202	0.465	0.298	<u>0.658</u>	0.468	0.817	<u>9.349</u>	17.869

Table 2: CRPS results on PhysioNet and Air Quality. Best results are in bold; second-best are underlined.

Method	PhysioNet			Air Quality
	10% missing	50% missing	90% missing	
GP-VAE	0.582	0.796	0.998	0.402
V-RIN	0.814	0.845	0.932	0.534
CSDI	0.242	0.336	0.528	0.108
CSBI	0.247	0.332	0.527	0.110
SSSD	0.233	0.331	0.522	0.107
TS-Diff	0.249	0.348	0.541	0.118
D ³ M	0.223	<u>0.327</u>	<u>0.520</u>	<u>0.106</u>
FSDI (Ours)	<u>0.229</u>	0.324	0.508	0.103

4.1.4 Evaluation Metrics. To evaluate the performance of FSDI, we report Root Mean Squared Error (RMSE), Mean Absolute Error (MAE) and the continuous ranked probability score (CRPS).

MAE. MAE estimates the average L_1 distance between ground truth and the imputation:

$$\text{MAE} = \frac{\sum_{k,\ell} m_{\text{tar}}^{k,\ell} |\hat{x}_{k,\ell} - x_{k,\ell}|}{\sum_{k,\ell} m_{\text{tar}}^{k,\ell}}. \quad (29)$$

RMSE. RMSE is the square root of the average L_2 distance between ground truth and the imputation results:

$$\text{RMSE} = \sqrt{\frac{\sum_{k,\ell} m_{\text{tar}}^{k,\ell} (\hat{x}_{k,\ell} - x_{k,\ell})^2}{\sum_{k,\ell} m_{\text{tar}}^{k,\ell}}}. \quad (30)$$

CRPS. CRPS measures the compatibility of an estimated probability distribution F with an observation t , which is formulated as:

$$\text{CRPS}(F^{-1}, t) = \int_0^1 2\Lambda_\alpha(F^{-1}(\alpha), t) d\alpha, \quad (31)$$

where $\Lambda_\alpha(q, z) := (\alpha - \mathbb{I}_{z < q})(z - q)$ and $\mathbb{I}$ is an indicator function. Following prior work [41, 45], we generated 100 samples to

approximate the distribution F . Then we can approximate CRPS by

$$\text{CRPS}(F^{-1}, t) \simeq \frac{1}{19} \sum_{i=1}^{19} 2\Lambda_{0.05*i}(F^{-1}(0.05*i), t). \quad (32)$$

So we finally report the average CRPS for all features and time steps:

$$\text{CRPS}(F^{-1}, x) = \frac{\sum_{k,\ell} m_{\text{tar}}^{k,\ell} \text{CRPS}_{k,\ell}(F_{k,\ell}^{-1}, x_{k,\ell})}{\sum_{k,\ell} m_{\text{tar}}^{k,\ell}}. \quad (33)$$

4.2 Time Series Imputation Results

Table. 1 reports the imputation performance in terms of MAE and RMSE on the PhysioNet and Air Quality datasets under different missing rates. On PhysioNet, FSDI achieves the best or second-best results across all settings, indicating that the proposed method can maintain strong reconstruction quality even when the observed ratio decreases to 10%. On the Air Quality dataset, FSDI also shows competitive performance in both MAE and RMSE. Overall, these results confirm that FSDI achieves stable and accurate imputation performance across different datasets.

Table. 2 presents the CRPS on the PhysioNet and Air Quality datasets. Compared to other probabilistic imputation methods, FSDI consistently ranks among the best across most settings. On PhysioNet, it achieves the lowest CRPS at 50% and 90% missing ratios, and remains competitive at 10% missing. On the Air Quality dataset, FSDI also attains the best performance, showing its ability to model capability of the distribution of unknown points. These results demonstrate that FSDI not only improves deterministic imputation accuracy but also enhances the quality of probabilistic imputation.

Figure 3 shows the visualization results on the AQI dataset and Physionet dataset. We can see that almost all ground truth values for the points to be imputed fall within the 95% confidence interval, and most of the ground truth values are within the 50% confidence interval, which demonstrates the effectiveness of FSDI.

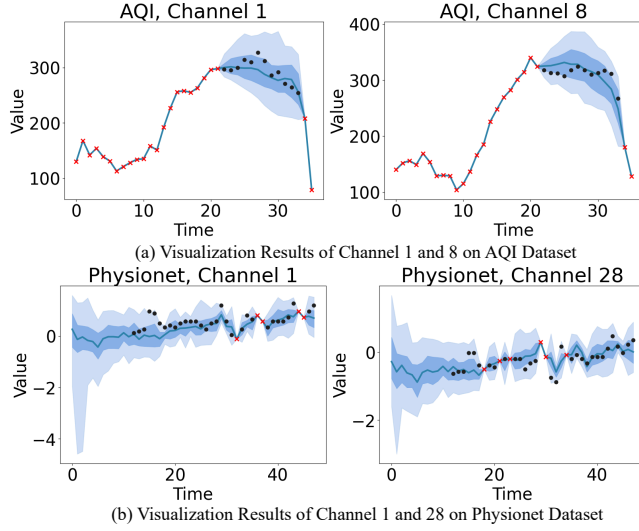


Figure 3: Visualization results of probabilistic time series imputation results on AQI and PhysioNet dataset. The green solid line represents the mean of the imputation results, while the dark blue represents the 50% confidence interval, and the light blue solid line represents the 95% confidence interval. The black dots represents the target of imputation and the red crosses represent the observed values.

4.3 Ablation Studies

To validate the effectiveness of the proposed methods, we conduct ablation experiments on PhysioNet with a 90% missing ratio. As introduced in Sec. 3, FSDI introduces two modifications over CSDI [41]: (i) the uniform variance schedule across frequency bins is replaced by a data-driven per-frequency-bin variance schedule via Frequency Importance Estimation (FIE), and (ii) an additional frequency-domain loss is introduced to encourage spectral consistency during training. Apart from these changes, all other components, including the backbone network and conditioning, remain identical to CSDI, ensuring a controlled comparison. We therefore evaluate four configurations: (i) CSDI: the original CSDI setup with a uniform per-frequency-bin variance schedule and no frequency-domain loss, (ii) w/o *FIE*: FSDI with the uniform variance schedule across frequency bins while retaining the frequency-domain loss, (iii) w/o $\mathcal{L}_{\text{freq}}$: FSDI without the frequency-domain loss and (iv) FSDI.

The results are shown in Table. 3. It can be observed that FSDI with the data-driven per-frequency-bin variance schedule already achieves notable improvements over CSDI across all metrics, indicating that reallocating noise across frequency bins is the primary driver of improvement. Introducing the frequency-domain loss provides further gains, although the margin is smaller, suggesting that it acts as an auxiliary regularization that enforces spectral consistency. Overall, the complete FSDI model yields the best performance, demonstrating the complementary benefits of the variance schedule and the frequency-domain loss.

Table 3: Ablation studies on PhysioNet with 90% missing ratio. Best results are in bold; second-best are underlined.

Method	MAE	RMSE	CRPS
CSDI	0.481	0.834	0.528
w/o <i>FIE</i>	0.479	0.831	0.520
w/o $\mathcal{L}_{\text{freq}}$	<u>0.475</u>	<u>0.826</u>	<u>0.515</u>
FSDI	0.468	0.817	0.508

Table 4: Parameter sensitivity results of λ_{freq} on PhysioNet with 90% missing. Best results are in bold.

λ_{freq}	MAE	RMSE	CRPS
0	0.475	0.826	0.515
0.01	0.473	0.825	0.514
0.05	0.468	0.817	0.508
0.10	0.470	0.823	0.510
0.20	0.470	0.823	0.510
0.50	0.468	0.819	0.510

4.4 Parameter Sensitivity Analysis

Table. 4 studies the sensitivity of the frequency-domain loss weight λ_{freq} on PhysioNet under the 90% missing setting. We observe that introducing a small but positive weight improves MAE, RMSE, and CRPS compared with the case $\lambda_{\text{freq}} = 0$, indicating that the spectral regularization contributes to more consistent imputations. Performance peaks around $\lambda_{\text{freq}} = 0.05$, which achieves the lowest errors across all three metrics. When λ_{freq} is further increased, the results remain competitive but don't show additional gains, and very large values tend to stabilize rather than improve performance. Overall, the analysis suggests that FSDI is not highly sensitive to the exact choice of λ_{freq} in a moderate range, while a small weight such as 0.05 is adopted as the default configuration in our experiments.

5 Conclusion

In this paper, we proposed FSDI, a frequency-shaped diffusion framework for time series imputation. By replacing the uniform variance schedule with a data-driven per-frequency-bin variance schedule that is Parseval-calibrated, FSDI reallocates the noise budget across the frequency bins while preserving the standard diffusion energy. Additionally, a lightweight frequency-domain loss encourages spectral consistency during training without introducing extra parameters into the sampler. Experiments on PhysioNet and Air Quality show that FSDI delivers strong and stable performance under a range of missing ratios, improving deterministic accuracy and enhancing probabilistic quality.

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A Experimental Details

A.1 Hyperparameters

Table 5 lists the hyperparameters in FSDI.

Table 5: Hyperparameter details in our experiments.

Dataset	Physionet	Air Quality
Epochs	200	200
Batch Size	16	16
Learning Rate	0.001	0.001
Layers	4	4
Channels	64	64
Heads	8	8
Diffusion Embedding Dim	128	128
β_0	0.0001	0.0001
β_1	0.5	0.5
λ_{freq}	0.05	0.05
Steps	50	50
Noise Scheduler	quad	quad

A.2 Additional Ablation Studies

To investigate the contribution of the Parseval constraint on noise allocation, we performed an ablation study where we removed this

constraint from FSDI. The results are shown in Table 6, indicating that the Parseval constraint has a small but noticeable positive effect on performance, especially on AQI.

Table 6: Additional ablation study results.

Dataset	Method	MAE	RMSE	CRPS
PhysioNet	FSDI	0.468	0.817	0.508
	w/o Parseval	0.470	0.815	0.510
AQI	FSDI	9.349	17.869	0.103
	w/o Parseval	9.669	19.737	0.109

A.3 Efficiency

We evaluate the efficiency of both CSDI and FSDI in terms of training and inference time on the PhysioNet dataset. The results, shown in Table 7, reveal negligible overhead.

Table 7: Training and inference time for CSDI and FSDI on PhysioNet.

Method	Training (s/epoch)	Inference (s/sample)
CSDI	9.0	0.048
FSDI	9.0	0.050

A.4 Downstream Performance

To assess the downstream impact of our method, we conducted a forecasting experiment on AQI using TimeGPT in a zero-shot setting. The results, shown in Table 8, demonstrate that using better imputation strategies leads to improved downstream performance. Specifically, FSDI imputation results in the best performance, confirming that improvements in imputation methods directly enhance downstream forecasting.

Table 8: Downstream performance (forecasting) on AQI using TimeGPT. All methods are evaluated in a zero-shot setting.

Method	MAE
No Imputation	0.4141
CSDI Imputation	0.2961
FSDI Imputation	0.2876

B Visualization Results

The imputation results on Air Quality dataset and Physionet2012 dataset with missing ratio 10%, 50% and 90% are presented in Fig.4,5,6,7.

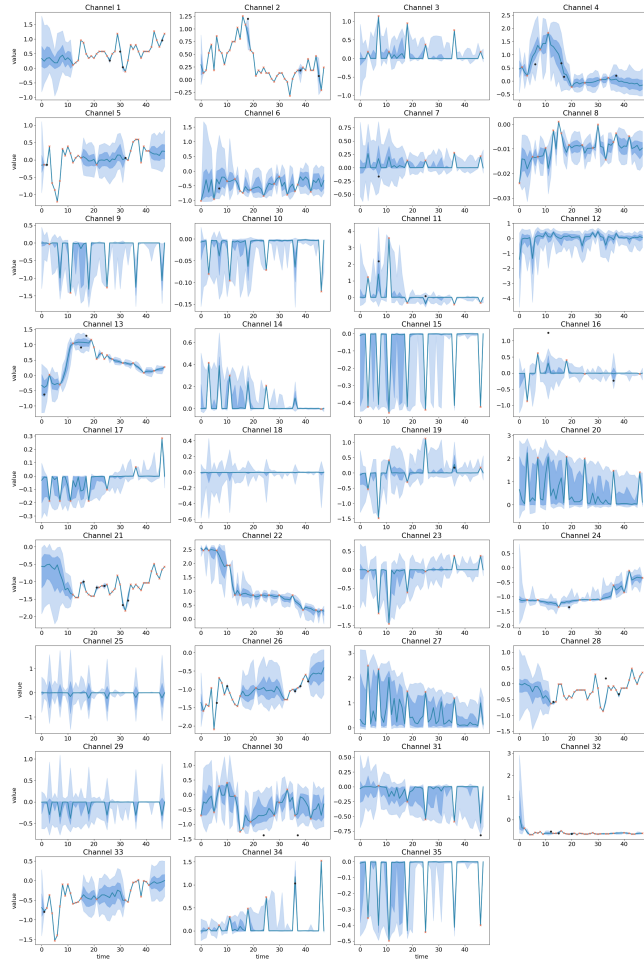


Figure 4: Visualization of imputation results on Physionet2012 dataset from Channel 1 to Channel 35 with 10% missing. The green solid line represents the mean of the imputation results, while the dark blue represents the 50% confidence interval, and the light blue solid line represents the 95% confidence interval. The black dots represents the target of imputation and the red crosses represent the observed values.

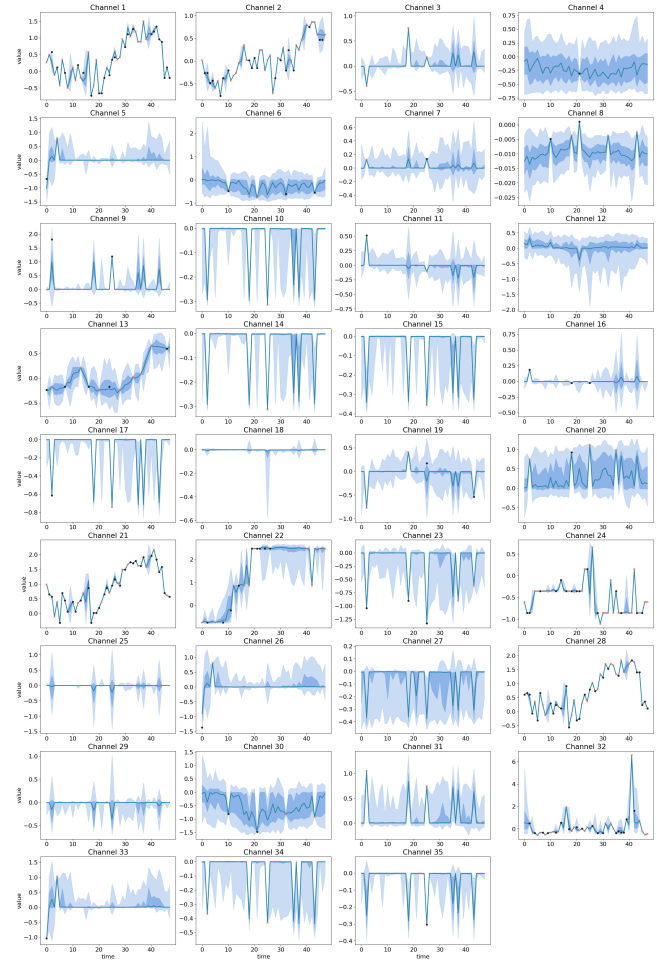


Figure 5: Visualization of imputation results on Physionet2012 dataset from Channel 1 to Channel 35 with 50% missing. The green solid line represents the mean of the imputation results, while the dark blue represents the 50% confidence interval, and the light blue solid line represents the 95% confidence interval. The black dots represents the target of imputation and the red crosses represent the observed values.

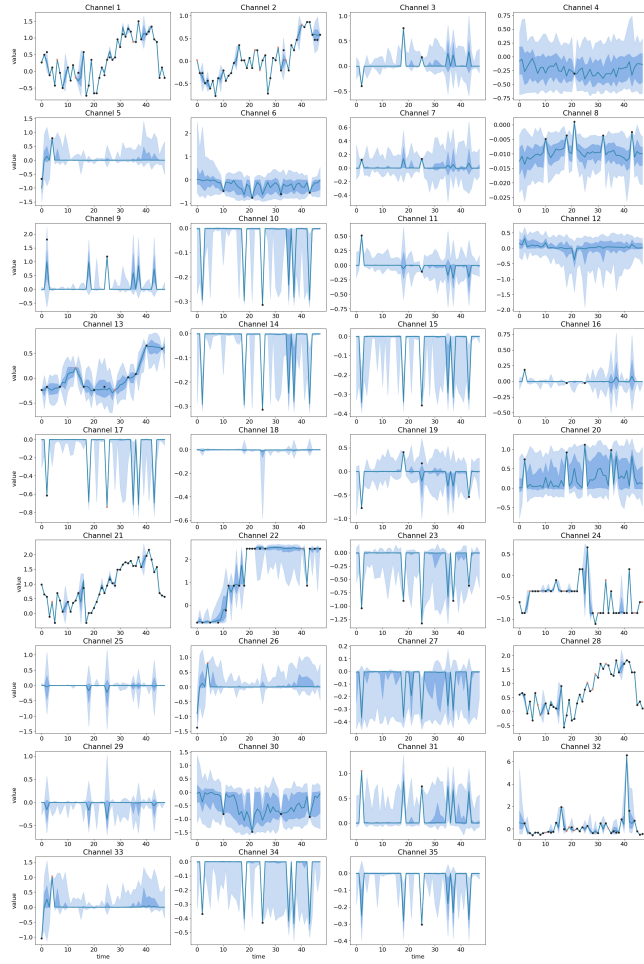


Figure 6: Visualization of imputation results on Physionet2012 dataset from Channel 1 to Channel 35 with 90% missing. The green solid line represents the mean of the imputation results, while the dark blue represents the 50% confidence interval, and the light blue solid line represents the 95% confidence interval. The black dots represents the target of imputation and the red crosses represent the observed values.

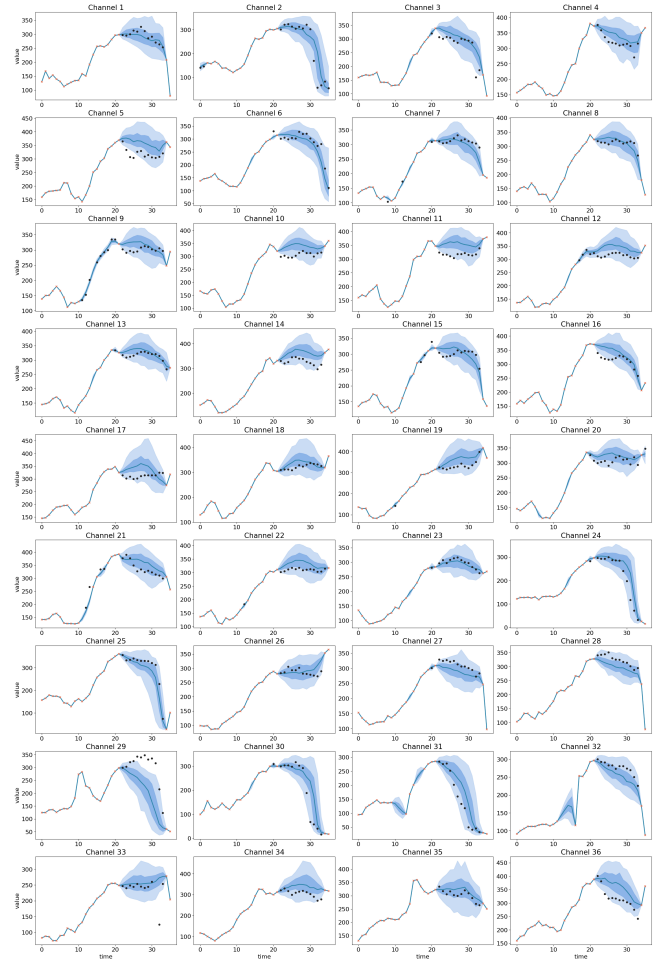


Figure 7: Visualization of imputation results on AQI dataset from Channel 1 to Channel 36. The green solid line represents the mean of the imputation results, while the dark blue represents the 50% confidence interval, and the light blue solid line represents the 95% confidence interval. The black dots represents the target of imputation and the red crosses represent the observed values.